Mir Usama Ul Haq



Test Validation & Verification Engineer | Infotainment & ADAS (HiL, Test Automation)

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Work permit: EU Blue Card – authorized to work in Germany, no sponsorship required

PROFILE

Test Validation & Verification Engineer with 3+ years of experience in automotive infotainment validation for BMW and hardware-in-the-loop (HiL) testing of ADAS functions at Bosch. Skilled in test case design, test automation (ECU-TEST, pytest, CANoe, vTESTstudio/CAPL, Python), HiL/SiL testing, SOME/IP, UDS, DLT/Ethernet trace analysis and test rig commissioning. Created 200+ ECU-TEST test cases and automated a complete Log & Trace test set, cutting weekly manual testing of 3–4 days to a few hours. Experienced in Scrum with Jira and Confluence. M.Eng. in Software-based Automotive Systems. EU Blue Card holder with unrestricted work authorization in Germany. English C1, German B1 (B2 in preparation).

EXPERIENCE

Automotive Infotainment Test Engineer May 2023 – Present
In-Tech GmbH, Munich, Germany | Client project: BMW Group (Log & Trace, infotainment validation)

- Designed, documented and executed 200+ test cases in ECU-TEST for automated infotainment software validation; automated the complete Log & Trace test set, reducing weekly regression effort from 3–4 days of manual testing to a few hours per run.
- Evaluate test runs through ECU-TEST graphical reports and failure-cause analysis to speed up defect triage.
- Analyze DLT and Ethernet traces (Carmen, DLT Viewer) to isolate defects.
- Apply pytest, SOME/IP and UDS in infotainment validation (test scripting, service communication, diagnostics).
- Provide support for LogTrace and measurement technology (Xoraya Loggers & Capture Modules) to BMW validation teams and BMW-linked partner companies worldwide in the Information and Communication domain.
- Commission and maintain 6 test racks; repair and assemble wiring harnesses and connectors to keep rig hardware operational; trained 7–8 new employees in rack setup.
- Manage project tickets in ALM Octane and Jira; maintain the customer-side LogTrace Confluence pages.
- Work in a Scrum team.

Intern – ADAS (Advanced Driver-Assistance System) Integration Oct 2022 – Apr 2023
Robert Bosch GmbH, Leonberg, Germany | Projects: Toyota, Ford

- Implemented hardware-in-the-loop (HiL) tests for driver-assistance systems and operated the HiL test bench.
- Developed new test cases and debugged existing ones in vTESTstudio using CAPL.
- Integrated radar and camera sensors into the HiL setup.
- Built restbus simulations for CAN networks in CANoe.
- Developed Python scripts to integrate hardware into automated test execution.
- Worked in a Scrum-based integration team.

Embedded Systems Software Developer Jun 2020 – Jul 2021
Gravixar, Islamabad, Pakistan

- Built an IoT agriculture monitoring system (ESP32/ESP8266 microcontrollers, C++ firmware, Microsoft Azure database).
- Upgraded legacy industrial automation with new sensors, actuators and C++ embedded algorithms.

EDUCATION

Master of Engineering – Software-based Automotive Systems Sep 2021 – Aug 2025
Hochschule Esslingen University of Applied Sciences, Esslingen, Germany
Final grade: 2.8 (German scale) | Full-time program

Master's Thesis – Simulation Environment for Testing Connected Cars Mar 2025 – Aug 2025
Completed alongside full-time employment at In-Tech GmbH

Bachelor of Science – Mechatronics Engineering Nov 2016 – Jun 2020
University of Engineering and Technology (UET) Taxila, Pakistan
Final grade: CGPA 3.5/4.0 (approx. 1.8 on the German scale, modified Bavarian formula)

PROJECTS

Master's Thesis – Simulation Environment for Testing Connected Cars Mar 2025 – Aug 2025

A web-based simulation platform was developed to test application-layer interactions in connected vehicles:

- **FastAPI** Python backend for command validation, state management, overrides, network delays/faults, timeouts, and vehicle-type-specific rules (EV, HEV, ICE, FCEV).
- Two synchronized, role-separated frontends (HTML/CSS/JavaScript) – Vehicle Manager (operator) and Vehicle Dashboard (end-user) – for real-time command issuance and feedback.
- Simulated remote commands (lock/unlock, preconditioning, status updates) with edge-case handling and synchronization between both frontends, on a modular, lightweight software-in-the-loop (SiL) architecture that enables rapid prototyping without hardware dependency.

Semester Project – Data Protection Quality in VANETs Mar 2022 – Aug 2022

Developed simulations and a V2X application/stack to evaluate the privacy of V2X applications:

- Traffic simulation (SUMO) and communication simulation (OMNeT++), synchronized together.
- Implementation of V2X privacy mechanisms.

Bachelor's Thesis – Autonomous Navigation of a Mobile Robot (ROS) Sep 2019 – Jun 2020

Autonomous robot developed for quick, reliable, and efficient package transport within an industrial setting:

- 3D model and electrical circuit designed in SolidWorks and Proteus.
- Object detection with LIDAR; localization and mapping via SLAM (Python).
- Real-time simulation in ROS Gazebo; optimized path selection using A* in Python.

SKILLS

Test & Validation: Test automation • Test case design • HiL • SiL • ECU-TEST • pytest • vTESTstudio • CANoe • CANalyzer

Protocols & Trace Analysis: CAN • LIN • Ethernet • SOME/IP • UDS • DLT (DLT Viewer, Carmen) • Xoraya Suite • Wireshark

Programming: Python • C • C++ • Embedded C • CAPL

Tools & Process: Scrum (Agile) • Jira • Confluence • ALM Octane • Git • Linux • MATLAB/Simulink • ROS • Visual Studio

Standards: ISO 26262 • ASPICE • AUTOSAR

CERTIFICATIONS

ISTQB Certified Tester Foundation Level – in progress

Automotive Embedded Systems & Applications (Udemy) Feb 2022

- AUTOSAR, RTOS, ASPICE, ISO 26262; CAN, LIN, TTL, FlexRay; wireless networking, Bluetooth security and BLE.
- Project: autonomous search-and-rescue robot (IR, ultrasonic and RGB-color sensors, magnetometer); C++ control algorithm on Arduino.

Mastering CAN Networks: Vector CANoe, CANalyzer, CAPL (Udemy) Jul 2022

- CAN protocol fundamentals; Vector hardware and software configuration; CANoe, CAPL, and CANalyzer in practice.

LANGUAGES

German B1 (B2 in preparation) • English C1 • Urdu (Native)